

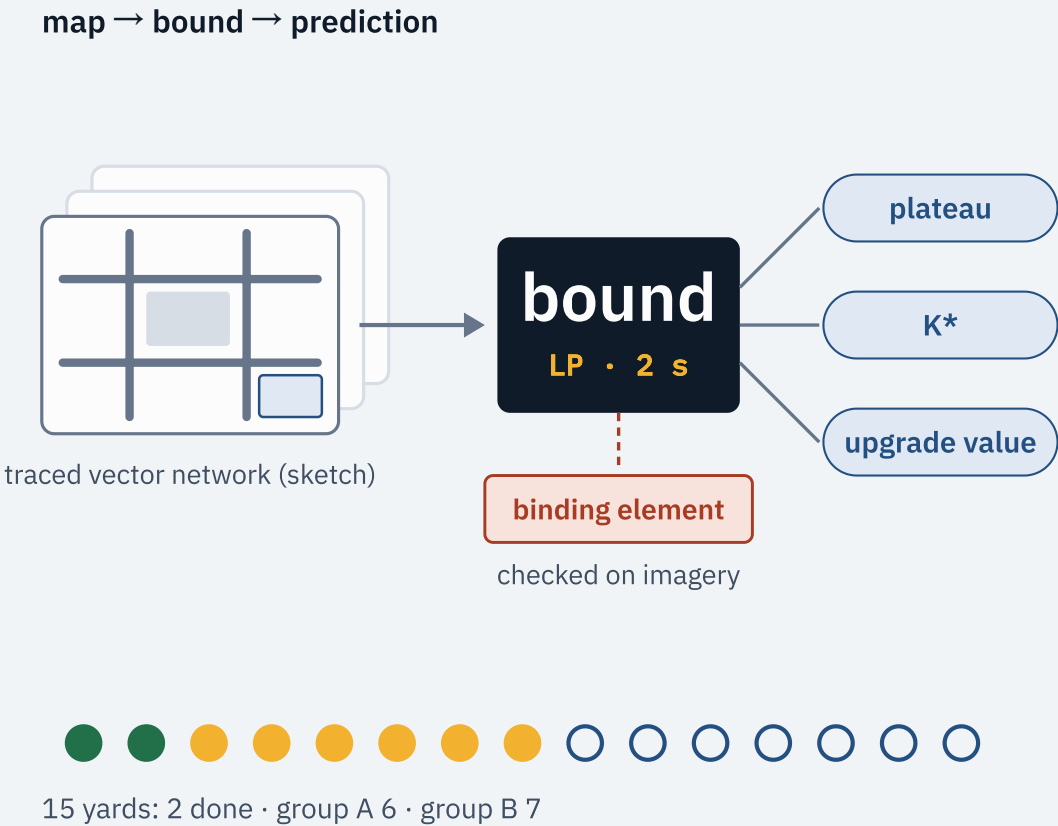
One map, a two-second LP: can it predict a new shipyard's capacity?

Testing across shipyards: the network bound computed from the map, with its binding elements, as a cross-yard predictor (transfer across yards; cross-yard prediction hypothesis)

Do jamming thresholds transfer across shipyards? Network bounds and binding elements as predictors on public layouts (working title)

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2 yards done · group A's 6 being annotated · 15 in all



This sub-study in one minute

1

The question

Do capacity results from two yards carry over? Chapter 6 tests the claim that the map-computed bound and its binding elements predict plateaus, jamming thresholds and upgrade values across yards.

2

The test

The predictor is the bound itself, a two-second LP per yard, with its binding elements; tested by leave-one-out, yards grouped by key element. Topological indicators explain where the bound binds; they do not replace it.

3

Two points so far

Okpo is a dock-road yard, Yantai a door yard. Reservation plateaus are 36% and 35% of the system bound; each predicts the other within about 3%. At own volume Yantai needs 1–2 vehicles; Okpo's peak (16 h) needs 117: network-bound.

4

Next

All 20 listed yards have maps and base maps; group A's six are being annotated. Per yard: bound and binding-resource reading, imagery check of the top three, then simulations. Group B follows at eight yards; 15 in all.

Bound first, binding element next, simulation last: a two-second LP screens a new yard.

The question: which yards do the earlier results carry over to?

5	Transfer across yards The network bound computed directly from the map, and its binding elements, predict plateaus, jamming thresholds and upgrade values across shipyards
The cross-yard prediction hypothesis	
Hypothesis	The network bound and binding elements computed directly from the map predict the simulated plateau, K^* and upgrade values across shipyards. Topological indicators serve to explain why the bound binds where it does.
Test	Leave-one-out test; classification by key element: dock-road single-lane type, door type, ...
If it holds	A two-second linear program is enough to screen a new shipyard.
If it fails	What transfers is the bound tool and the key-element classification themselves.

Where chapter 6 sits in the thesis

Ch. 3 Capacity	The network bound, the capacity interval [best deadlock-free rule, network bound] and the K^* criterion: the predictor and the first two targets
Ch. 4 Orchestration	Pushes the interval's lower edge up: plateaus move with the rule, so each rule is read separately
Ch. 5 Co-design	Dual prices and upgrade simulations: the third target, upgrade value
Ch. 6 Many yards	Which yards these results carry over to: prediction errors, the key-element classification and a screening procedure

The hypothesis pays either way: if it holds we gain a screening tool for new yards; if not, the bound tool and the key-element classification are themselves what transfers.

The bound itself predicts, not a topology regression on 15 yards

Regressing topological indicators on 15 yards

- Fifteen samples support one or two predictors at most; easy to overfit
- Traffic rules and task flows sit between indicators and capacity

The network bound as the predictor

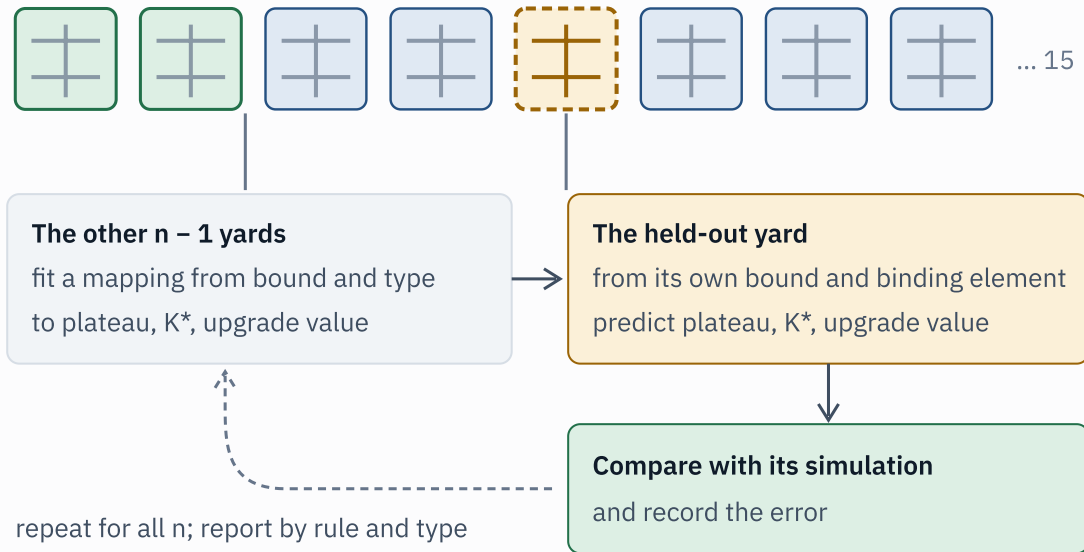
- From the map in two seconds per yard: an occupancy LP with width, crane and window rows
- System bound: the task stream's expected occupancy, no sampling noise (Okpo 1,736, Yantai 1,298 a day)
- Upper edge of the interval (two ceilings) under conditions 1–3, loaded routes fixed; the route-free bound holds for any routing
- Dual prices name the binding element to check first on imagery
- Structurally among the closest: railway capacity without a timetable (Jensen et al. 2020, branch and bound)

Topology of the two yards: to explain, not to replace

	OKPO	YANTAI
Roads / centreline	721 / 40.3 km	140 / 8.6 km
Junctions / entrances / stops	334 / 218 / 421	58 / 32 / 140
Independent cycles μ / meshedness α	82 / 0.064	14 / 0.056
Dock-bound minimum cut (sum of widths)	62 m	137 m
Dock entry degree (max)	5	5
→ system bound (a day)	1,736	1,298
→ binding element	dock-road single-lane	door type
→ $K^*(20\%)$	50	25
→ reservation plateau (a day)	627	455

Similar loop density, yet K^* differs twofold and so does the binding element. Indicators explain why: a narrow dock-bound cut at Okpo, a door at Yantai.

The test: leave-one-out, with yards grouped by key element



Two OD samplings (by entity, by stop) are each yard's standard sensitivity. New types come from the new yards' dual prices, not fixed in advance; each type records the resource granularity it was read under.

Three targets

Plateau

The interval's lower edge, read per traffic rule (chapters 3, 4)

K^*

Network takes over: marginal gain first below 20% of free flow (network-bound test, chapter 3)

Upgrade value

Dual-price prediction against upgrade simulations (chapter 5; run after the rule comparison)

Two key-element types so far (a third reading on page 8)

Dock-road single-lane · Okpo

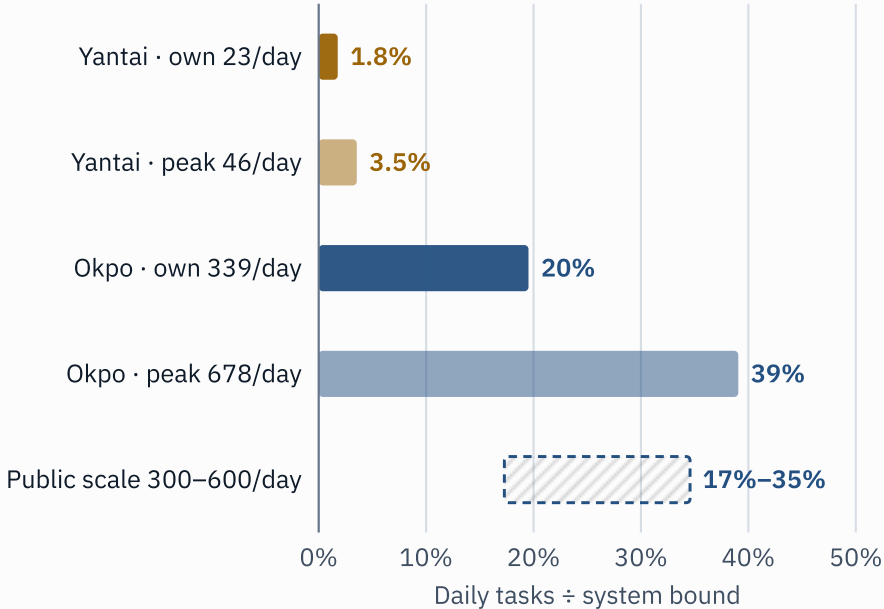
The dock-1 road, dock 1's only stopping road (9 m × 1,095 m); the bound binds there in 5 of 10 seeds, on a narrow through corridor in 4.

Door type · Yantai

The paint shop's two doors (presumed): each takes about 7% of all tasks and is held about 10 min per load; the bound binds there in 9 of 10 seeds.

First two data points: Okpo and Yantai, two key-element types

	OKPO	YANTAI
Map version (of these results)	before floating docks, spurs	after the second door
System bound (a day; 10-seed mean in brackets)	1,736 (1,720)	1,298 (1,252)
Capacity interval [best deadlock-free rule, bound]	[627, 1,720]	[455, 1,252]
Reservation plateau ÷ system bound	36%	35%
K*(20%) (range)	50 (50–60)	25 (25–25)
Where the bound binds (of 10 seeds)	dock-1 road ×5 · through corridor ×4 · 18 m road ×1	two paint-shop doors ×9 · another shop's door ×1
Type	dock-road single-lane	door type
Dock threshold (safe release, whole road, K = 20 → 150)	0.98 → 0.59	0.93 → 0.38
Own daily tasks ÷ system bound (mean–peak)	339–678: 20%–39%	23–46: 1.8%–3.5%
Vehicles needed at own mean (16 h, reservation) vs K*	27 < 50	1 < 25

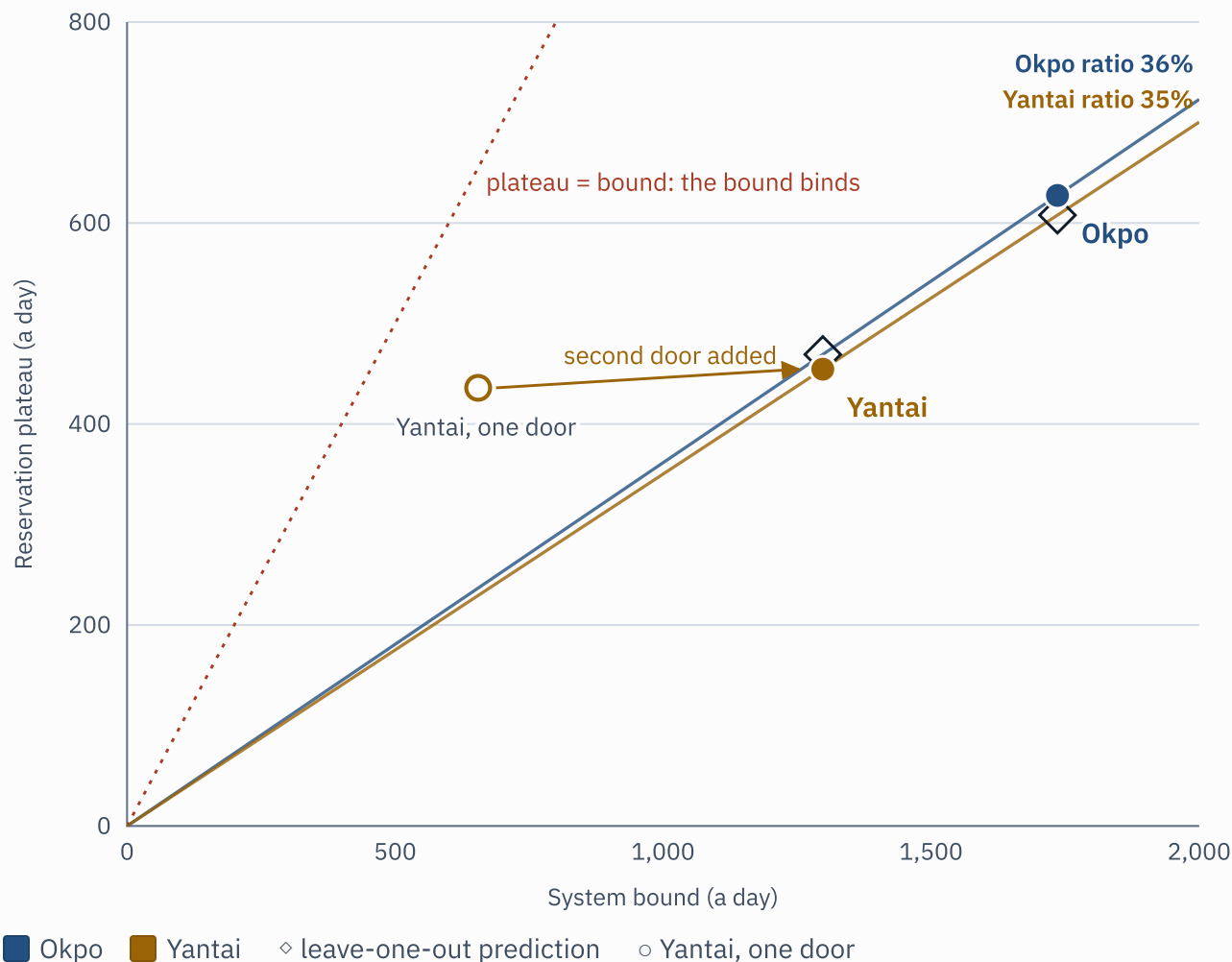


■ Yantai ■ Okpo
▨ public scale of large Korean yards, not Okpo itself (Yim 2008; Shen et al. 2018)

Density criterion: daily tasks ÷ system bound (network-bound hypothesis).
One folding for both yards: steel ÷ 100 t × 10 moves ÷ 300 days. Okpo 1,018,067 t a year → 339 a day (294–363 across years; Hanwha's direct anchor 350); Yantai 68,512 t → 23 a day.

Sources: simulations in this study; Okpo's daily volume folded as in the external-anchor check (100 t blocks after Zhang 2025, 10 moves after Wang 2017); public scale 300–600 a day from Yim 2008, Shen et al. 2018. Conditions: whole-route reservation (best deadlock-free rule tested), saturated, fleet-sizing paper's flow mix, plateau = mean over K = 100–150, whole-segment resources, 10 seeds; thresholds: page 9.

First leave-one-out: plateaus at 36% and 35% of the system bound



36% / 35%

Reservation plateau ÷ system bound: Okpo / Yantai

±3.2%

Error when one yard's ratio predicts the other's plateau (±0.5% with the 10-seed sampled bound)

67% → 36%

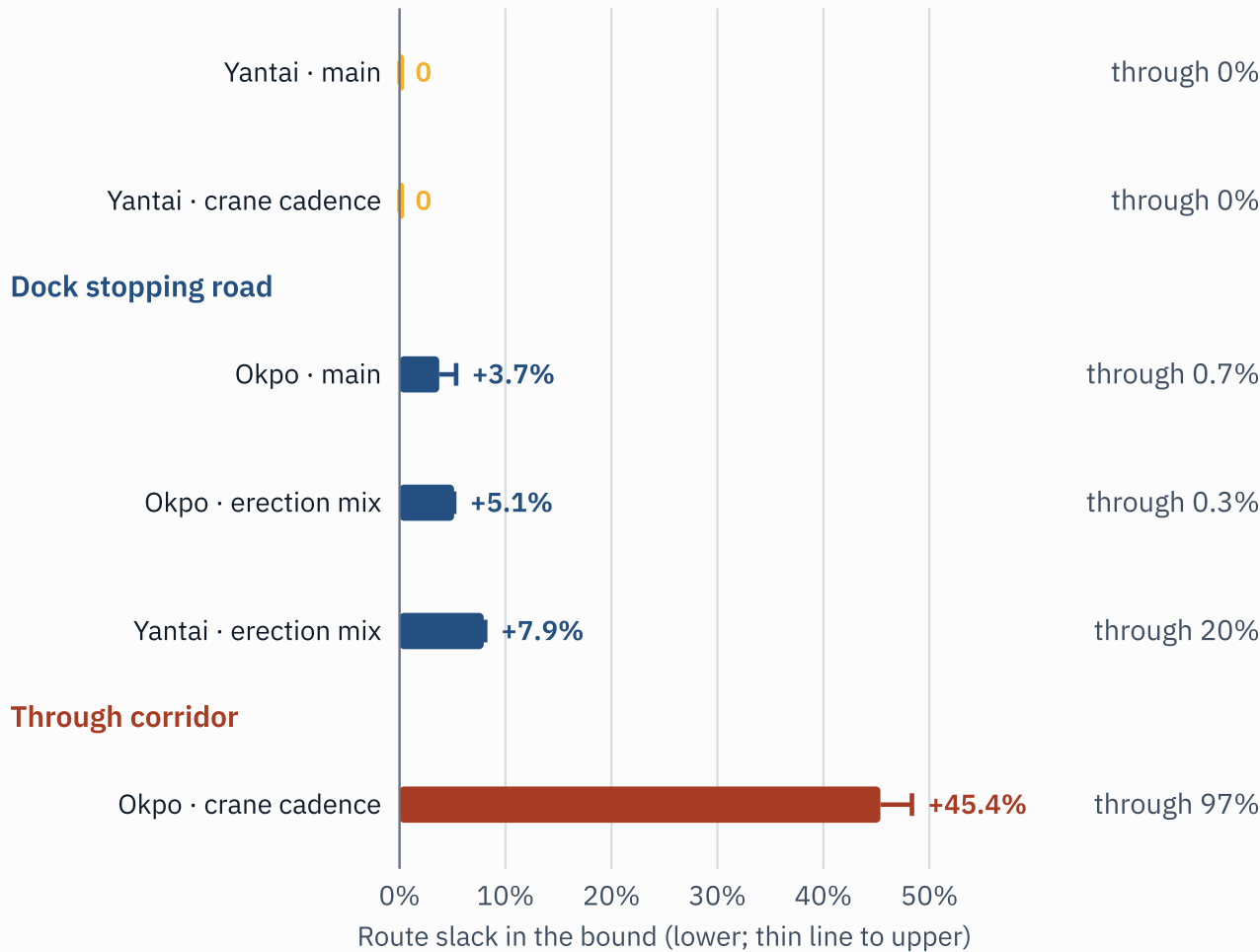
Yantai before and after the paint shop's second door (10-seed mean):
bound 655 → 1,252, plateau 436 → 455

Two points only show how the test runs. The predictor is the system bound: the sampled bound runs low in expectation (a draft Jensen proposition), and without that noise the error is about 3%. The ratio moves with the traffic rule, so chapter 6 reads it per rule and per type. Near 1 the bound binds and road changes pay; far below 1, orchestration first: Okpo's erection mix, 0.87 and 0.53 with blocks, marks the boundary between traffic ordering and the road-vehicle exchange rate.

Sources: simulations in this study: system bounds, 10-seed means of the sampled bound, reservation plateaus (mean over K = 100–150); Yantai one-door: 10-seed mean on the map before the second door; erection-mix ratios from the physical-rule runs. Leave-one-out: one yard's plateau ÷ system bound × the other's system bound. Conditions as on page 6; the boundary is part of the exchange-rate claim.

The binding resource, read three ways: measurable type signatures

Entrance



Entrance · door type

Yantai's paint-shop doors: no through traffic, zero slack by construction; every route is the same. Only the door moves the bound (retrofit runs: door granularity).

Dock stopping road · dock-road type

Okpo's dock-1 road, Yantai's dock stopping road: 0.3–20% through traffic, slack +3.7% to +7.9%, from stops entered at the other end. Granularity, such as blocks, is the first-order lever.

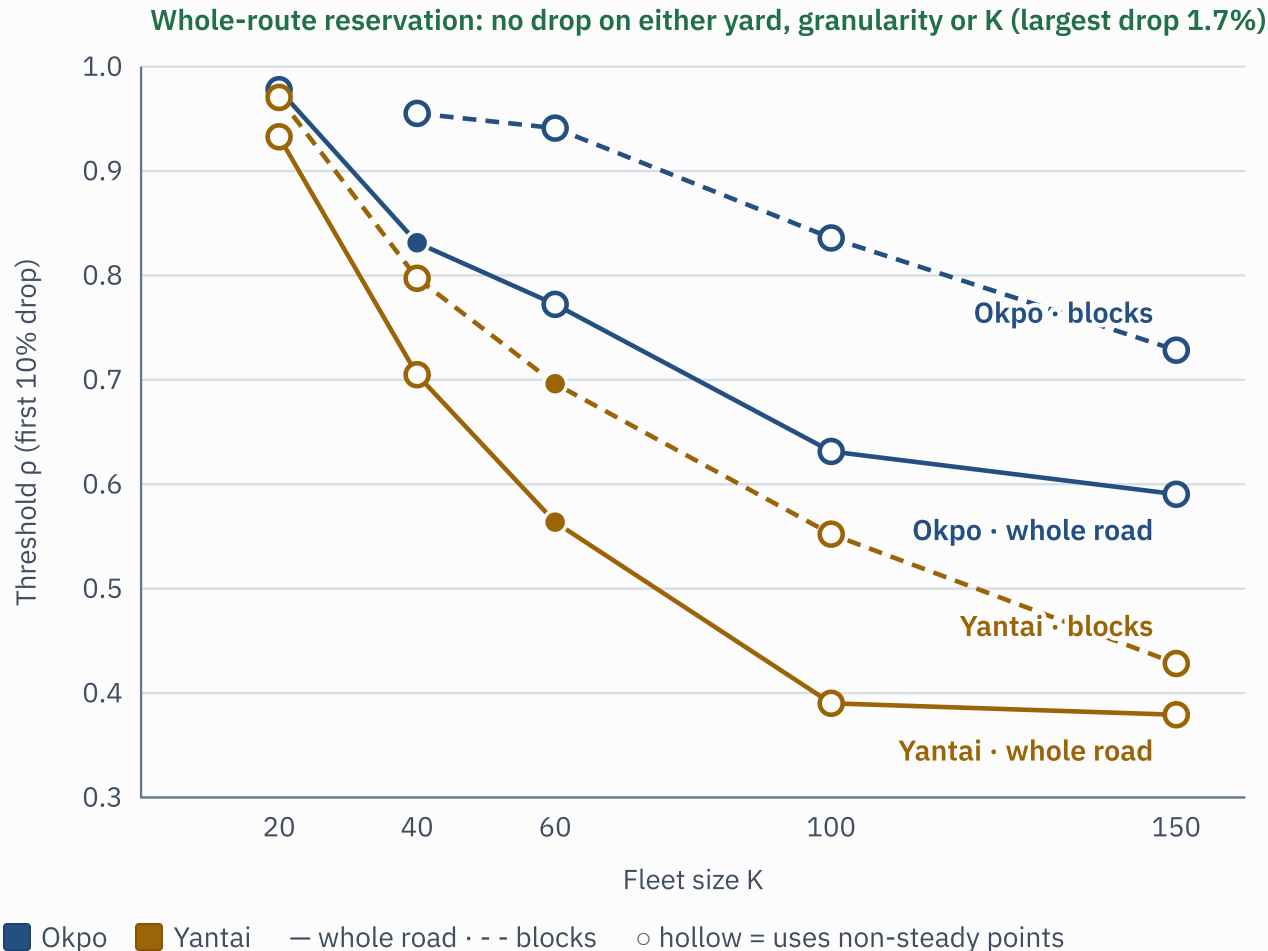
Through corridor · newly read

An Okpo corridor under crane cadence: 97% through traffic; its detours keep 22% spare and add 235 m; slack +45.4% (upper +48.4%). Detour orchestration can recover it (chapter 4).

Both signatures (through share, spare occupancy on the detour) come from the bound's solution, without simulation, and define the cross-yard types. Six scenarios so far, 1–3 points per type; upgrade scenarios add more.

Sources: simulations in this study. Conditions: system bound (loaded moves on shortest paths), route-free bound (any route), steady state, saturated, Okpo map before floating docks and spurs, Yantai map. Route slack = difference of bounds, not an achievable gain (lower: back-substituted feasible value – network bound; upper: route-free – network bound); no counterexample to the route-free bound found.

Second target: the dock threshold, same mechanism, yard-specific values



1

Wait off the road: no drop

Whole-route reservation holds on both yards, both granularities, $K = 20-150$ and p up to 0.99 (largest drop 1.7%).

2

Wait on the road: the threshold moves left with fleet size

Safe release (no teleport), whole road, $K = 20 \rightarrow 150$: Okpo $0.98 \rightarrow 0.59$, Yantai $0.93 \rightarrow 0.38$; blocks move it right (Okpo, $K = 40$: $0.83 \rightarrow 0.96$).

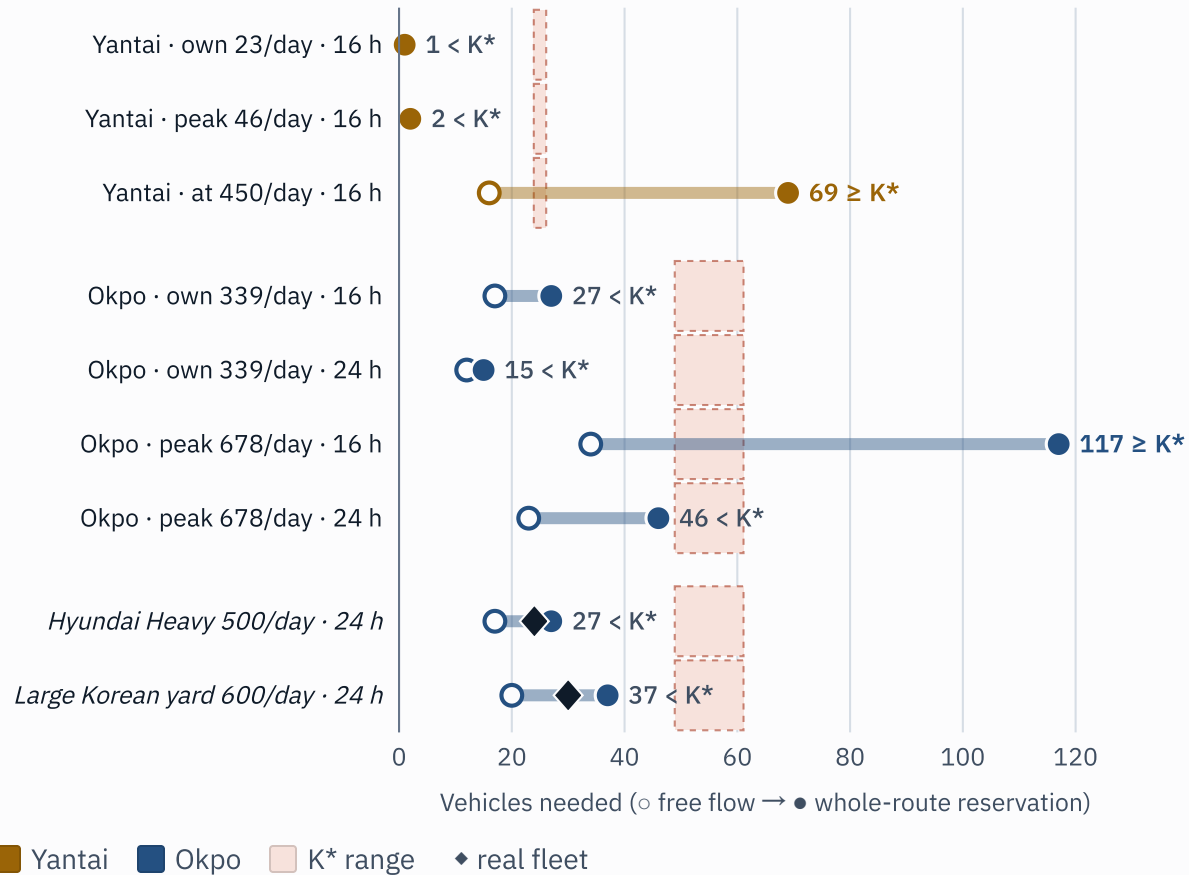
3

The mechanism transfers; the values must be predicted

At every K Okpo's threshold is above Yantai's ($K = 60$: $0.77 / 0.56$). Chapter 6 has to get this number from each yard's key elements; the three-resource dock network of the retrofit runs relates the threshold to the passable room at the dock, a candidate predictor.

Sources: simulations in this study (physical rules and whole-yard volumes). Conditions: crane-cadence p curves, saturated, 8 simulated days (day 1 warm-up), 10 seeds; threshold = the p at which throughput first drops 10% below the lowest p step (Okpo 0.53, Yantai 0.13), interpolated between steps; hollow = uses non-steady points; Okpo blocks at $K = 20$ never drop 10% and are not drawn; Okpo map before floating docks and spurs, Yantai map. Safe release = segment requests with a banker's-algorithm-style safety check, deadlock-free, no teleport.

Either side of the network-bound regime: vehicles needed at whole-yard volumes against K^*



1

Yantai: its own volume is on the other side

Its own 23 / 46 a day need 1–2 vehicles, far below $K^* = 25$; at 450 a day (16 h) it needs 69 and enters the regime too.

2

Okpo: below at its mean, above at its peak

Own 339 a day: reservation needs 27 (16 h) or 15 (24 h), below K^* (50–60). Peak 678: 117 at 16 h, inside the regime; 46 at 24 h, at its lower edge.

3

Real fleets sit between the model's two ends

Hyundai Heavy's 24 vehicles (about 500 a day, 24 h) and a large Korean yard's about 30 (about 600 a day) both sit between free flow and reservation under a 24 h day.

The main caliber (16 h or 24 h, mean or peak) is set after the deadlock-free rule comparison; both are shown here.

Sources: simulations in this study (physical rules and whole-yard volumes; Yantai's own volume from the crane-cadence and dock-mouth runs). Conditions: main flow mix, whole-segment resources, 4 days, 10 seeds; vehicles needed = smallest integer K whose 10-seed mean throughput $\geq 99\%$ of the daily tasks; K^* range Okpo 50–60, Yantai 25; peak = mean $\times 2$. Field anchors: 박창규 & 서준용 2006 (Hyundai Heavy, 24 h); Shen et al. 2018 (yard unnamed, shifts not stated, read at 24 h); both checked in full text.

Key-element review: the top three elements, checked on imagery

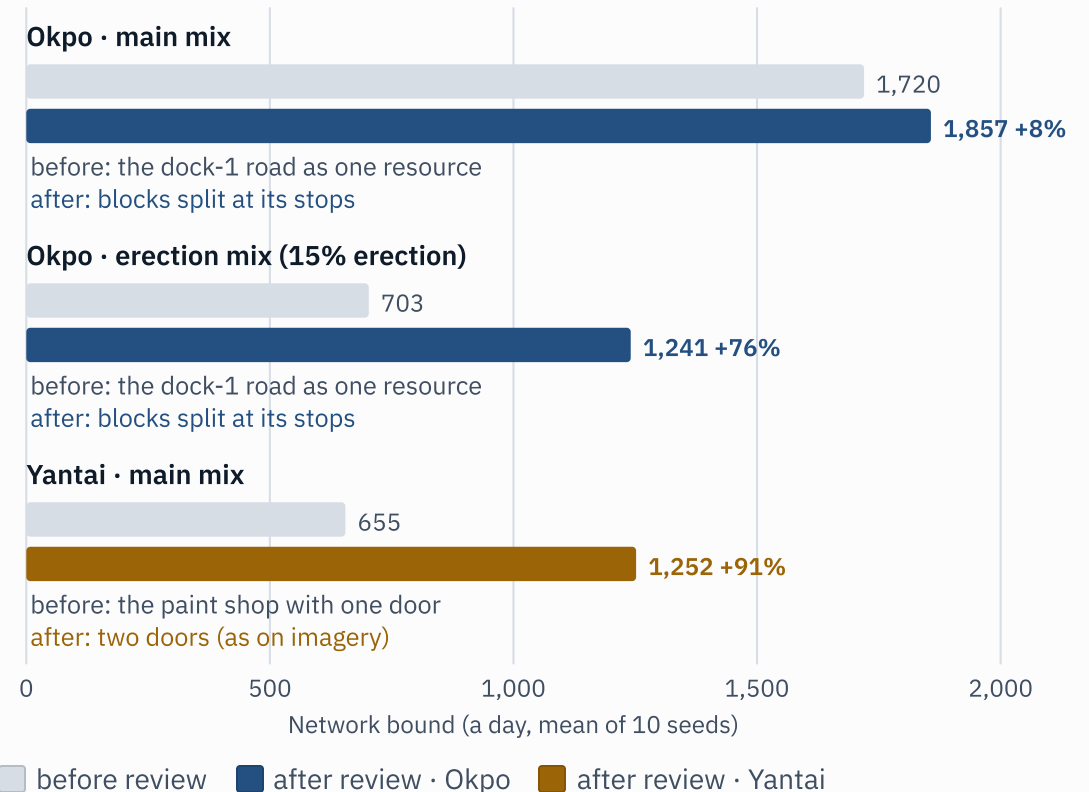
- 1 List the top three elements by dual price**, with building, entrance name and coordinates, not just a label
- 2 Check three things on the imagery**: road width; the number of doors; how traffic passes roadside stops and doors (while a vehicle works there, can others get in and out, can they pass?)
- 3 Record it in the map's source field**
- 4 Choose the resource granularity and recompute the bound**: whole road or blocks, door held or shared

Okpo · dock-1 road

The model held the 1.1 km dock road as one resource. Imagery: neighbouring assembly sites stay reachable while a vehicle works at dock 1. → split at the stops: main mix upper edge +8%, lower +0.3%; erection mix upper +76%, lower +8.3%. Blocks are Okpo's main scenario.

Yantai · paint shop

At first one door was mapped; the review added the second, as on the imagery. On 9 Oct: vehicles can pass at the doors, so holding a door while loading is conservative. → the retrofit runs (after the rule comparison) add door granularity.



On both yards the binding element sat on a conservative modelling choice, and the check changed the model. Three elements per yard, no precise annotation: a minimal human check, written into the thesis as an engineering method.

Fifteen yards: 2 done, 6 in group A next, 7 in group B

— Done — Group A: next, reaching eight yards in six countries — Group B: up to 15

Hanwha Ocean Okpo KR LNG carriers, container ships, offshore Flag-experiment yard; dock-road type. Submarine and naval work disclosed	CIMC Raffles Yantai CN rigs, FPSOs, wind installation vessels Offshore yard; door type. Public environmental and cleaner-production filings	Chantiers de l'Atlantique FR cruise ships Large European cruise yard; IGN open imagery, the richest OSM	Meyer Werft Papenburg DE cruise ships Covered building docks, a distinct dock type; LGLN open imagery	Imabari Saijo JP bulk and ore carriers, container ships A building dock over 400 m with three large gantry cranes; GSI imagery
Philly Shipyard US Jones Act tankers, container ships Small single-dock yard; NAIP public-domain imagery	New Yangzi Shipbuilding CN container ships, bulk carriers, tankers Large multi-dock yard; reports of a Singapore-listed group	Samsung Heavy Geoeje KR LNG carriers, container ships, offshore On the same island as Okpo; DART filings	Fincantieri Monfalcone IT cruise ships Second European cruise yard, civilian only; Esri 2025 imagery	Oshima Shipbuilding JP bulk carriers Mid-size Japanese bulk-carrier yard; GSI imagery
Namura Imari JP bulk carriers, tankers A third Japanese yard; GSI imagery	CM Jinling Weihai CN ro-pax ferries Small yard with a distinct product	Hengli Heavy Industry CN bulk carriers, tankers, container ships Expanded in 2024; imagery must postdate it	HD Hyundai Samho KR VLCCs, container ships, LNG carriers Large yard; DART filings	K Shipbuilding KR MR tankers, mid-size container ships Mid-size yard; moved up from group C on 10 Oct, replacing HD Hyundai Mipo

Selection looks for spread: dock type, key-element type and the level of daily tasks ÷ bound matter more than the count.

Source: candidate yard list (10 Oct 2026); all 20 listed yards (2 done, 18 in groups A, B, C) have maps and base maps in the study's yard-map collection; list and reasons checked against public sources; flagged entries are verified with facility use before inclusion. Country codes: KR Korea, CN China, FR France, DE Germany, JP Japan, US United States, IT Italy.

Selection rules: civilian yards, open data first, wide variety

- 1 **Civilian yards:** naval builders left out (Hudong-Zhonghua, GSI Nansha, HD Hyundai Mipo); Okpo, the finished flag yard, stays, naval business disclosed
- 2 **Open data first:** open-licence orthophotos, else Esri imagery for local tracing; OSM roads as drafts; filings to check facility use and output
- 3 **Wide variety:** size, layout, dock type (open, covered, single, multiple), region, product
- 4 **Checks before inclusion:** imagery from 2022 on and after any expansion; blocks reach the dock by road
- 5 **Choose by key-element type and by daily tasks ÷ bound:** spread matters more than count

Two cautions

- **Philly Shipyard:** Hanwha plans expansion and naval orders; civilian for now on dated pre-expansion imagery, swapped for group C if it turns mainly naval
- **Covered dock halls:** if blocks do not reach the dock by road, rule 4 removes the yard

Group C reserves (5), for any A or B yard that fails a check

- New Times (Jiangsu) CN
- Shanghai Waigaoqiao CN
- JMU Ariake JP
- Tsuneishi Zhoushan (SW Xiushan) CN
- Daehan Shipbuilding KR

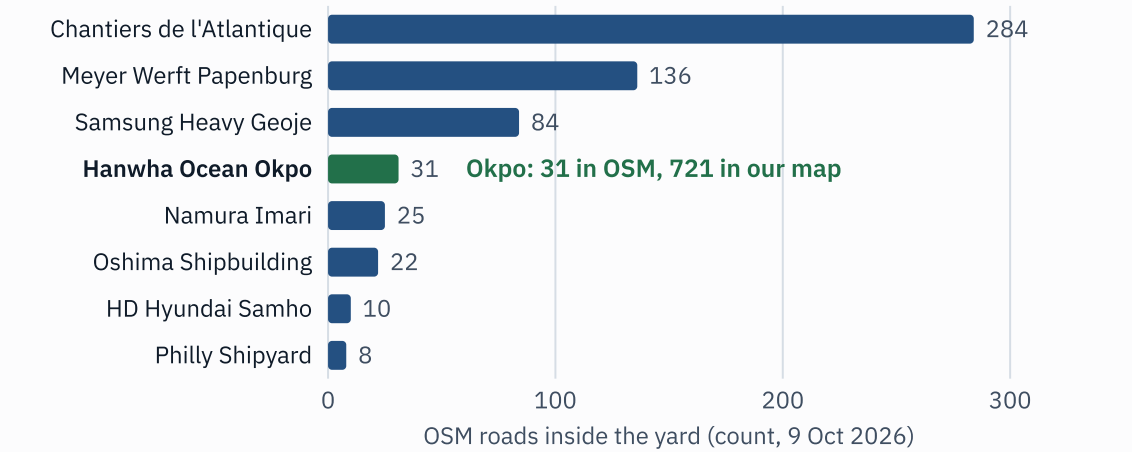
Removed from the list

- **Hudong-Zhonghua, GSI Nansha:** naval shipbuilding
- **Meyer Turku (Finland):** some blocks come by barge from Germany, not by yard road
- **HD Hyundai Mipo (10 Oct):** merged into HD Hyundai Heavy, some docks turning to naval ships; out under rule 1, K Shipbuilding moves up

Open imagery and OSM drafts; figures show only our traced vectors

REGION	IMAGERY	LICENCE AND USE
France	IGN BD ORTHO (about 20 cm)	Licence Ouverte 2.0, credit IGN
Lower Saxony, Germany	LGLN DOP20 (20 cm)	CC BY 4.0, credit “LGLN (year)”
Japan	GSI aerial photo tiles	use with attribution
United States	USDA NAIP (about 60 cm)	public domain
Friuli Venezia Giulia, Italy	IRDAT / Eagle.fvg (2017–2020)	open licence, but older than 2022: not used
China, Korea (existing base maps)	existing TIFs (about 0.25 m)	tracing only, never in figures
Other Chinese and Korean yards, Monfalcone	Esri World Imagery (2022–2025, 0.31–0.5 m)	not open: local tracing only, not redistributed (decided 10 Oct 2026)

Open-licence terms are checked against the originals on inclusion (IGN, NAIP and GSI pending). Capture dates: Philly NAIP 10 May 2022; the Esri yards December 2022 to August 2025 by metadata; the rest pending. Figures show only our traced vectors.

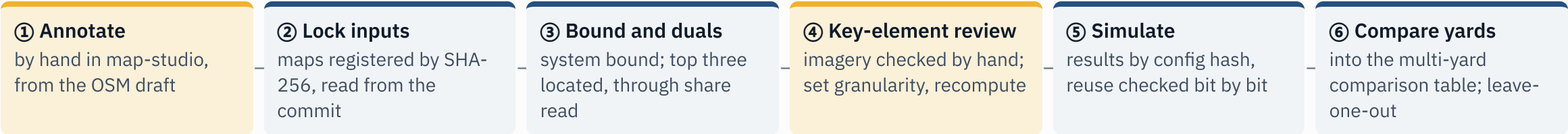


- **All 20 listed yards have maps and base maps** (the study's yard-map collection): 11 OSM drafts, 3 drafts by extent, 4 blank maps on existing base maps, copies for Okpo and Yantai; checked by hand in map-studio: roads added, widths measured, facilities and entrances marked
- **Data sharing:** OSM-derived parts are shared alike under ODbL
- **Optional accelerator:** the segmentation paper's model extracts roads, yards and buildings from imagery where OSM is sparse

Open the segmentation paper briefing →

Sources: candidate yard list (10 Oct 2026 version), imagery table and base maps per yard (OSM road counts from Overpass, 9 Oct 2026, © OpenStreetMap contributors, ODbL; Esri dates from World_Imagery_Metadata); imagery terms and ODbL per the licence texts. No imagery or network geometry is shown.

Per yard: bound first, then the key-element review, then simulation



STEP	WHO	STATUS	OUTPUT (CHAPTER 6)
Yard list; maps and base maps for 20 yards	curation	done	Table of yards and data sources
Annotating group A's 6 yards	manual	running	Multi-yard instance library (vector networks)
System bound, top-3 duals and binding-resource reading per yard; two OD samplings	computation	on each map	Comparison table: bound, binding element, type, through share
Key-element review, then recompute	manual	fixed step	Check notes (map source field)
Verify facility use and output; daily tasks by the same folding	curation, literature	per yard	Density chart; vehicles needed against K*
Flag simulations: best deadlock-free rule after the comparison, whole-yard calibers	computation	after the bound	Plateau, K*, capacity interval, dock threshold
Topology of 8 yards; order group B by the gaps	curation	after A	Indicator table that explains
Leave-one-out and types: a round at 8, final at 15	—	planned	Main figure: prediction against simulation; type table

Sources: selection steps of the candidate yard list; inputs locked by SHA-256, results by config hash, reuse checked bit by bit; chapter 6 plan (weeks 5–8 compute bounds and plateaus for the first yards of group A); yard maps always come from the study's yard-map collection (decided 10 Oct 2026).

What is established, and what is in progress

Established (two-yard simulations, literature checks)

- The network bound runs on both yards and no checked point exceeds it; the system bound removes sampling noise, the route-free bound holds for any routing
- Both ends of the capacity interval; the best deadlock-free rule is whole-route reservation
- Two key-element types, plus three readings of the binding resource
- Plateaus at about 36% / 35% of the system bound; the dock threshold works the same way on both yards
- Okpo's own 339 a day (external-anchor check); the two yards fall either side of the network-bound regime, real fleets between the model's two ends
- Key-element review is a fixed step; maps for 20 yards, SHA-256 locks, bit-by-bit reuse checks

TASK	CONTENT	WHAT IT BRINGS TO CHAPTER 6
Deadlock-free rules running	First, maps from the study's yard-map collection (Okpo after floating docks and spurs); a less conservative deadlock-free rule (a banker's-algorithm-style check plus a step-wise feasibility search, or claims cut at the next safe holding point, the safe-orchestration theorem); re-checks the lower edge and the decline and safe-orchestration-cost hypotheses	The rule for every yard's simulations; plateau ÷ bound may rise
Retrofit runs after the rules	On the new rule's code and the Okpo map with floating docks and spurs: upgrade simulations and the first road-vehicle exchange-rate table; door granularity; block-length curve; three-resource dock network	The upgrade-value target; more points for the three readings; a candidate threshold predictor
Floating docks queued	Floating-dock scenario; finite holding area S and $K_0(S)$	Okpo's final layout

After the rule comparison, three items are settled together: the network-bound hypothesis's new numbers (part 2), its two-layer wording (part 3), and the main caliber (16 h or 24 h, mean or peak).

Sources: simulations in this study (from the flag experiment to the system and route-free bounds; per-point bound checks); the external-anchor and literature check and the full-text comparison with the closest work; plans as of 10 Oct 2026 (the rule comparison was set first that day). Running and queued tasks are plans, not results.

Risks and mitigations

RISK	MITIGATION
Plateau ÷ bound moves with the rule and the key-element model (67% for Yantai with one door); a less conservative rule may lift the lower edge	Read per rule and type, one column per new rule; two OD samplings as standard sensitivity; review the key element first
The binding element sits on a conservative modelling choice (on both yards)	The key-element review is a fixed step; every reading states its granularity; through share and detour slack say which lever to change
Fifteen samples are few; each of the three readings has 1–3 points	No regression: the network bound as the predictor, types by key element, leave-one-out; upgrade scenarios add points on every yard
The network-bound reading depends on the working day and on mean or peak	16 h and 24 h, mean and peak side by side; the main caliber is set after the rule comparison; Hyundai Heavy and Shen et al. as field checks
Yards without suitable open imagery rely on Esri imagery; most capture dates still to check	Esri for local tracing only, never redistributed; figures show vectors only; terms and dates checked on inclusion; OSM parts under ODbL
A yard fails after selection (turns mainly naval, or blocks arrive by sea)	Five group C reserves; HD Hyundai Mipo already swapped for K Shipbuilding; Philly and covered halls have rules

Sources: candidate yard list (10 Oct 2026 version); simulations in this study (physical rules and whole-yard volumes; system and route-free bounds); the Yantai map before and after the second door.

Where paper four goes: journals in the school's 2026 list, and a working title

Working title (direction)

Do jamming thresholds transfer across shipyards? Network bounds and binding elements as predictors on public layouts

The original question, reframed around the cross-yard prediction hypothesis: the bound and its binding element as predictors

FIRST CHOICE · CLASS B

***Ocean Engineering* (class B)**

Engineering methods for shipyard production and transport; the 15-layout library and screening procedure are the contribution

***Journal of Marine Science and Application* (class B)**

Naval and ocean engineering; multi-yard instances and bound-based screening for practice

ALTERNATIVE · B STRETCH · A

***Engineering Applications of Artificial Intelligence* (class B)**

When joined with layout extraction by segmentation: imagery → vector network → bound → type

Class A

No natural class-A venue; not forced

SUPPLEMENTARY · CLASS C

***Journal of Ship Production and Design* (class C)**

Shipyard production organisation, the closest setting

Three more class-C venues: *Int. J. Naval Architecture and Ocean Engineering*, *Journal of Marine Science and Engineering*, *International Shipbuilding Progress*

Class B first, meeting the school's “at least one class-B-or-above journal paper”; no natural class-A venue, not forced.

References

1. Buhl, J., Gautrais, J., Reeves, N., Solé, R.V., Valverde, S., Kuntz, P., Theraulaz, G. (2006). Topological patterns in street networks of self-organized urban settlements. *The European Physical Journal B* 49(4), 513–522. doi:10.1140/epjb/e2006-00085-1
2. Fekete, S.P., Schepers, J. (2001). New classes of fast lower bounds for bin packing problems. *Mathematical Programming* 91(1), 11–31. doi:10.1007/s101070100243
3. Jensen, L.W., Schmidt, M., Nielsen, O.A. (2020). Determination of infrastructure capacity in railway networks without the need for a fixed timetable. *Transportation Research Part C* 119, 102751. doi:10.1016/j.trc.2020.102751
4. 박창규, 서준용 (2006). 조립블록 지번할당 알고리즘 개발: 현대중공업 사례. *IE Interfaces* 19(2), 160–167.
5. Shen, H., Jeong, Y.-K., Shin, J.G., Lee, P., Woo, J.H., Lee, Y.G., Kim, S.H., Jeong, J.H. (2018). Key performance indicators and analysis method for ship block logistics flow in shipyards. *Proc. 2018 Winter Simulation Conference*.
6. Stone, M. (1974). Cross-validatory choice and assessment of statistical predictions. *J. Royal Statistical Society B* 36(2), 111–133. doi:10.1111/j.2517-6161.1974.tb00994.x
7. Wang, C. 王冲 (2017). 船厂船体分段外场物流关键问题研究. 博士学位论文, 武汉理工大学.
8. Yim, S.-B., Roh, M.-I., Cha, J.-H., Lee, K.-Y. (2008). Optimal block transportation scheduling considering the minimization of the travel distance without overload of a transporter. *J. Society of Naval Architects of Korea* 45(6), 646–655. doi:10.3744/STAK.2008.45.6.646
9. Zhang, S. 张水根 (2025). 大型船海装备制造场地智慧调度与优化系统设计及实现. 硕士学位论文, 沈阳工业大学. doi:10.27322/d.cnki.gsgyu.2025.000182

Data and material: simulations in this study (results from the flag experiment to the system and route-free bounds) and this study's claims and hypotheses; the external-anchor and literature check; the candidate yard list (10 Oct 2026); the study's yard maps (traced from public imagery); OpenStreetMap contributors via Overpass (9 Oct 2026), ODbL 1.0; open imagery IGN BD ORTHO, LGLN DOP20, GSI tiles, USDA NAIP (terms checked on inclusion); Esri World Imagery (not open, local tracing only); CIMC Raffles cleaner-production notice (29 Apr 2024) and 2021 environmental disclosure; the school's 2026 journal list (23 Sep 2026).

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The predictor is the network bound itself

One two-second linear program per yard; reservation plateaus at 36% and 35% of the system bound, each predicting the other within about 3%; three readings of the binding resource.

Key-element review is a fixed step

After the bound, the top-three dual-price elements are checked on imagery before the resource granularity is set; in both yards the check changed the model.

Fifteen public layouts

Two done, on either side of the network-bound regime; all 20 listed yards have maps, group A's six being traced; grouped by key-element type and tested leave-one-out on plateau, K^* , threshold and upgrade value.

Bound first, binding element next, simulation last: chapter 6 tests whether a two-second linear program can screen a new shipyard.

[← Back to the thesis briefing \(road-occupying transport\)](#)[Occupancy model and bounds →](#)[Capacity interval →](#)[Fleet-network co-design →](#)[The fleet-sizing paper briefing →](#)[The segmentation paper briefing →](#)

Thank you.